DAY 24 | HANDS-ON ACTIVITIES

FDE Skills 3a and LLM Orchestration: Problem Statements

Three role plays (morning) and four technical problems (afternoon). Role play examples show what good looks like.

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How to work: read the problem, write the steps in plain English first, then write the code. Test with the sample run, then try your own inputs, including wrong ones.

Problem 1: Influence Without Authority

Level: Role play, pairs, 10 minutes | **Concepts:** influencing, framing

You need the customer's IT lead to open a firewall port for your pipeline. They are overloaded and wary of contractors. Prepare and run the conversation.

Sample run / expected result

Opens with their concern: "I know every change adds risk to your platform."
Offers evidence: a one-page diagram and the exact rule needed
Reduces effort: "I have drafted the change request for you to review"
Agrees a next step with a date

Hints

- Lead with their interest, not yours.
- Make saying yes easy.

Problem 2: The Deadline Negotiation

Level: Role play, pairs, 12 minutes | **Concepts:** interests, BATNA, trade-offs

The sponsor insists on all 8 features by the board meeting in 2 weeks. Your team can deliver 5 well. Find the interests and negotiate an agreement.

Sample run / expected result

Asks why the date matters (board wants proof of value)
Offers options: 5 features plus a demo of the next 3, or all 8 with 2 as prototypes
States the trade-off openly
Writes the agreement in a follow-up message

Hints

- Know your BATNA: what happens if there is no agreement?

Problem 3: Two Teams in Conflict

Level: Role play, groups of 3, 12 minutes | **Concepts:** conflict management, de-escalation

The data team blames the app team for bad numbers; the app team blames the data. You facilitate a call between them.

Sample run / expected result

Sets ground rules: facts first, no blame
Acknowledges frustration on both sides
Asks each side to describe the data flow
Agrees a joint check of one record end to end by tomorrow

Hints

- Move from people to process.
- Agree something small and concrete.

Problem 4: Summary Chain

Level: Easy | **Concepts:** ChatPromptTemplate, LCEL, batch

Build a chain that summarises a text for a given audience (CIO, engineer, end user) in 3 bullet points, and run it on one text for all three audiences with batch().

Sample run / expected result

```
CIO: cost and risk focus
Engineer: technical detail
End user: what changes for me
```

Hints

- prompt | model | StrOutputParser()

Problem 5: Structured Extractor

Level: Medium | **Concepts:** JSON output, validation

Extract employee, amount, currency and date from 5 free-text expense messages into validated Pydantic objects. Count how many fail validation.

Sample run / expected result

```
Expense(employee='Mei', amount=4200.0, currency='JPY', date='2026-11-20')
Valid: 5/5
```

Hints

- Validate every model output before using it.

Problem 6: RAG Chain and Router

Level: Challenging | **Concepts:** retriever runnable, RunnableBranch

Build a RAG chain over the handbook, then a router that sends HR questions to the RAG chain and small talk to a friendly reply chain.

Sample run / expected result

```
"When does leave expire?" -> rag -> 31 March
"Hello!" -> chat -> Hi! How can I help?
```

Hints

- Classify first with a cheap rule or a small prompt.

Problem 7: Brief a Coding Agent

Level: Medium | **Concepts:** AGENTS.md, task brief

Write an AGENTS.md for the Task Board repo and a task brief asking a coding agent to add a priority filter with tests.

Sample run / expected result

```
AGENTS.md: setup, test command, style rules, folders not to touch
Task: goal, files, acceptance criteria, tests to add, out of scope
```

Hints

- If a new colleague could not follow it, an agent cannot either.

Hanashiai (話し合い): talking things through together